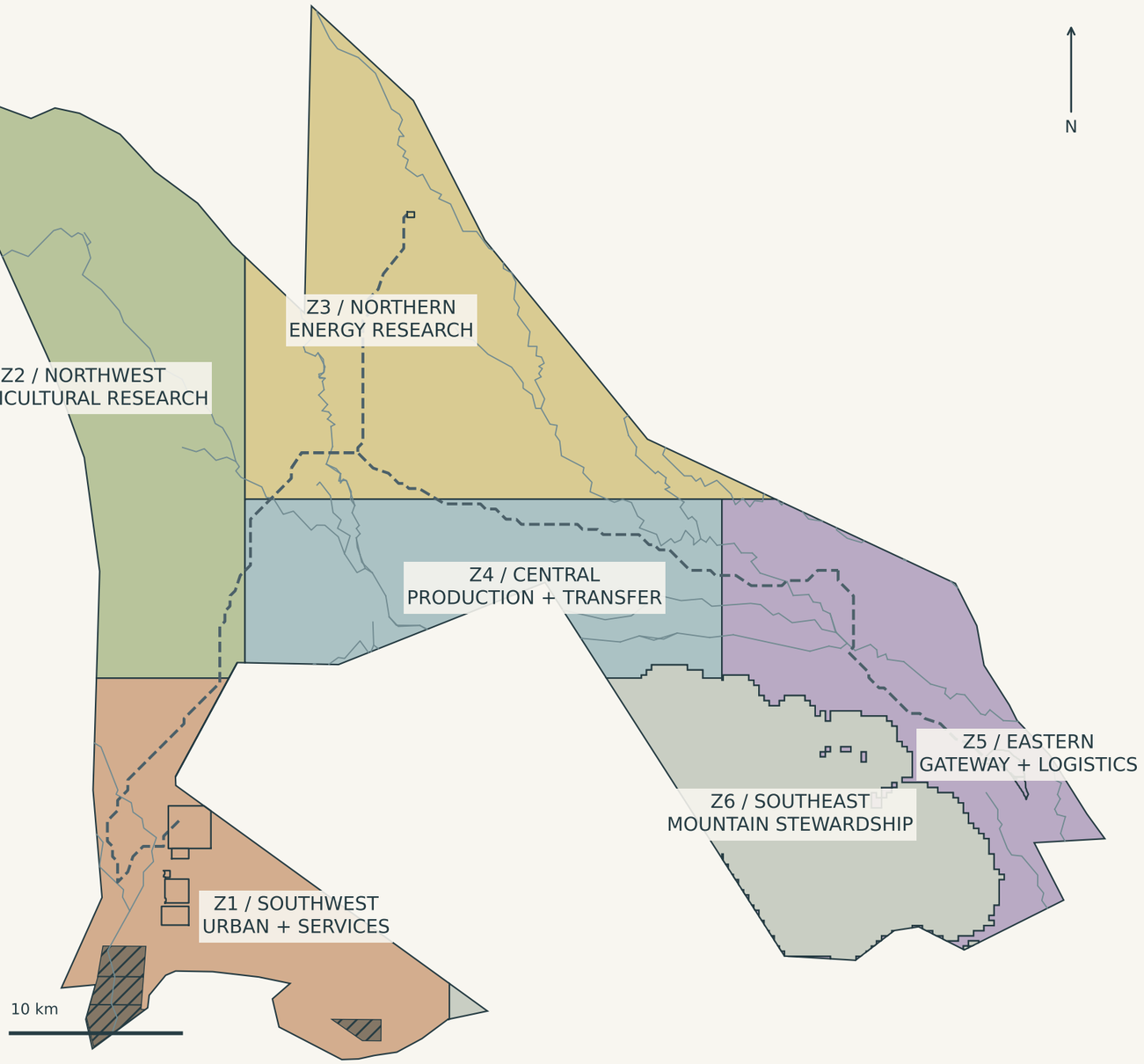
Whole-site regional plan / six coordinated regions

Strategic regions cover the entire issued site; construction land, ecological constraints and phases are separate layers.



Z1 / Southwest urban and service

232.6 km2 gross / proposed role

R + C continuous living district; separate P/S utilities; fixed pilots retained.

Z2 / Northwest agricultural research

341.2 km2 gross / proposed role

Dryland crop trials and agricultural service; no blanket irrigation or new town assumed.

Z3 / Northern energy research

392.3 km2 gross / proposed role

Energy research and northern maintenance hub; installations follow demand and power rights.

Z4 / Central production and transfer

238.2 km2 gross / proposed role

Regional transfer and conditional productive sites; source washes and terrain remain overlays.

Z5 / Eastern gateway and logistics

227.5 km2 gross / proposed role

Conditional gateway with worker services. Trade and access rights precede works.

Z6 / Southeast mountain stewardship

226.1 km2 gross / proposed role

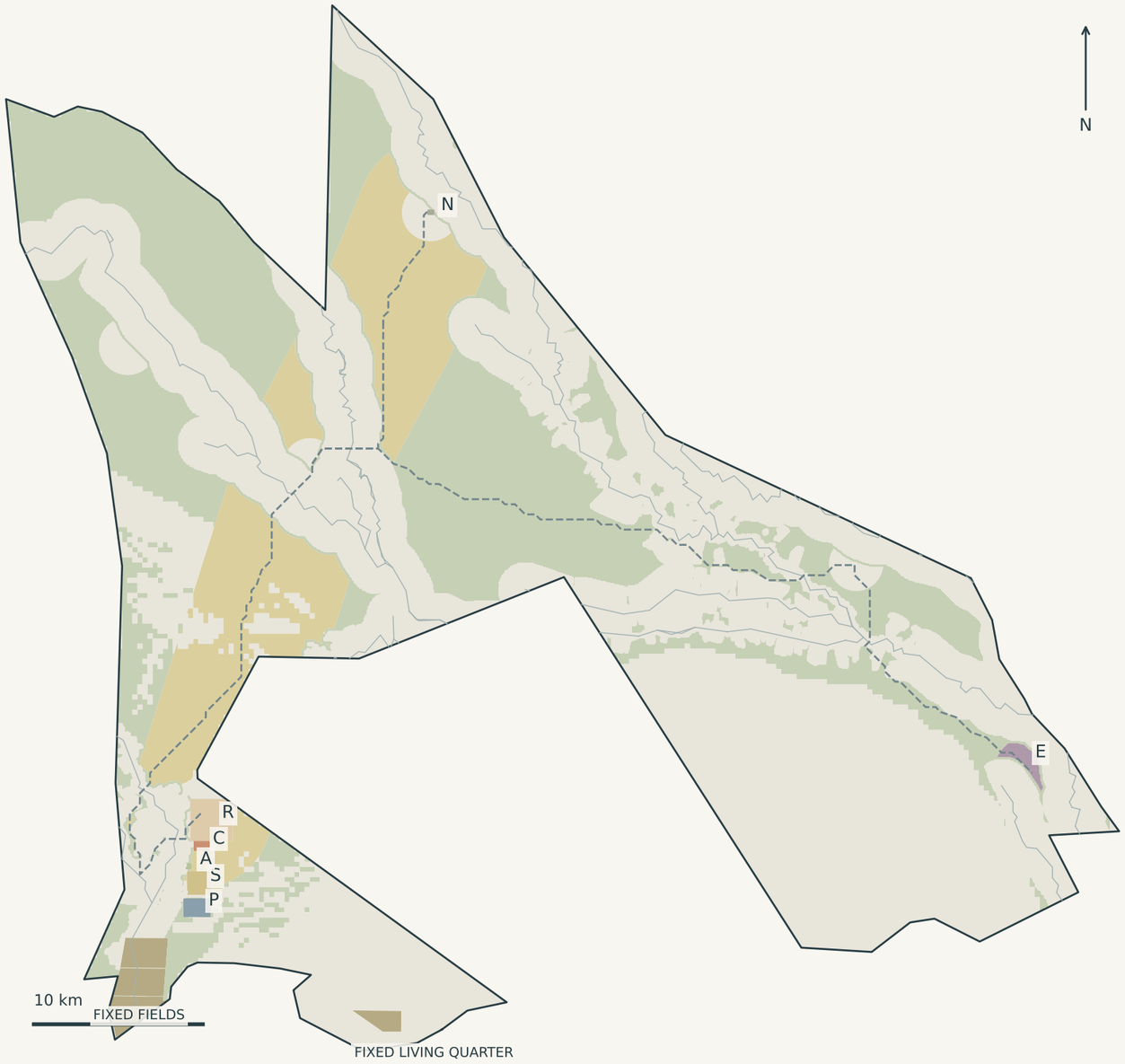
Mountain and dryland stewardship; no urban allocation. DEM terrain screening governs any access intervention.

Solid edges: proposed service-region demarcations (not statutory lines). Thin grey-blue: source seasonal channels. Dashed: regional corridor studies. Hatched: fixed pilots.

Z1 contains the detailed urban plan. Z2-Z6 remain region-scale proposals with stated activation gates; their gross areas are not approved buildable land.

The whole site and the complete proposed settlement system

Issued boundary retained / full planned construction is distinguished from territorial options and unbuilt land.



A complete city, built in three stages

10,240 designed dwellings / 25,600 resident scenario at full occupancy. All R/C blocks are assigned a phase and use. These are conditional design capacities, not observed demand.

Production at its own scale

P contains compute, repair and metered utility works. S contains explicit phased PV fields. N is a farm-service hub. E is a gateway workplace with roster beds, not a second residential city.

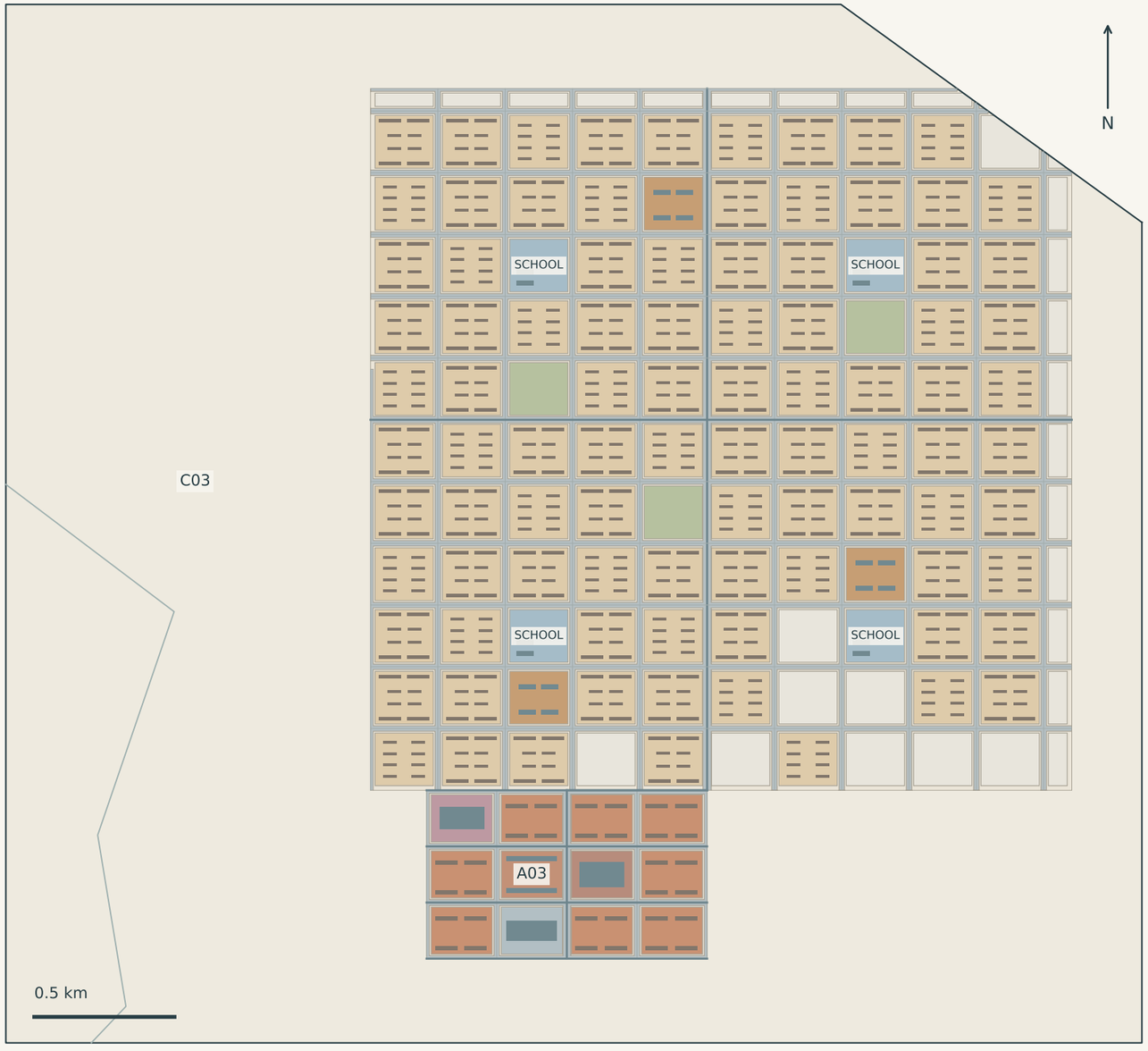
Territorial land is not all construction

Remaining energy/agriculture roles are options, not installed PV or irrigated fields. No added population is assigned to them. Fixed pilots are unchanged; forage grass and jojoba stay under their arrays.

Full-site map / current geometric area 1,658.82 km2; nominal competition label 1,656 km2 / dashed regional corridors are studies, not secured roads.

Complete urban layout and its productive counterpart

Readable blocks, streets, heights and anchors across all three phases; the productive inset is geographically separate.



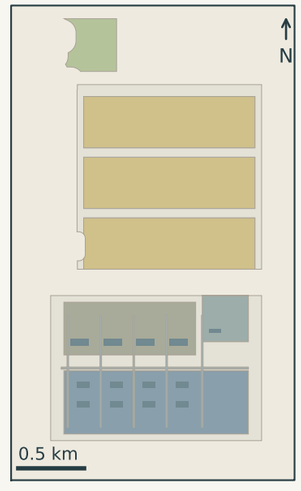
R / continuous residential fabric

91 housing blocks. Three-storey street-front apartments and two-storey terraces share public courts and streets. Terrain and boundary remnants are explicitly unbuilt, rather than counted as housing.

C / mixed-use centre at arrival

A compact market/commons, primary-health and learning buildings, passenger arrival and phased mixed-use blocks. Local schools and daily services remain in R; C does not replace them.

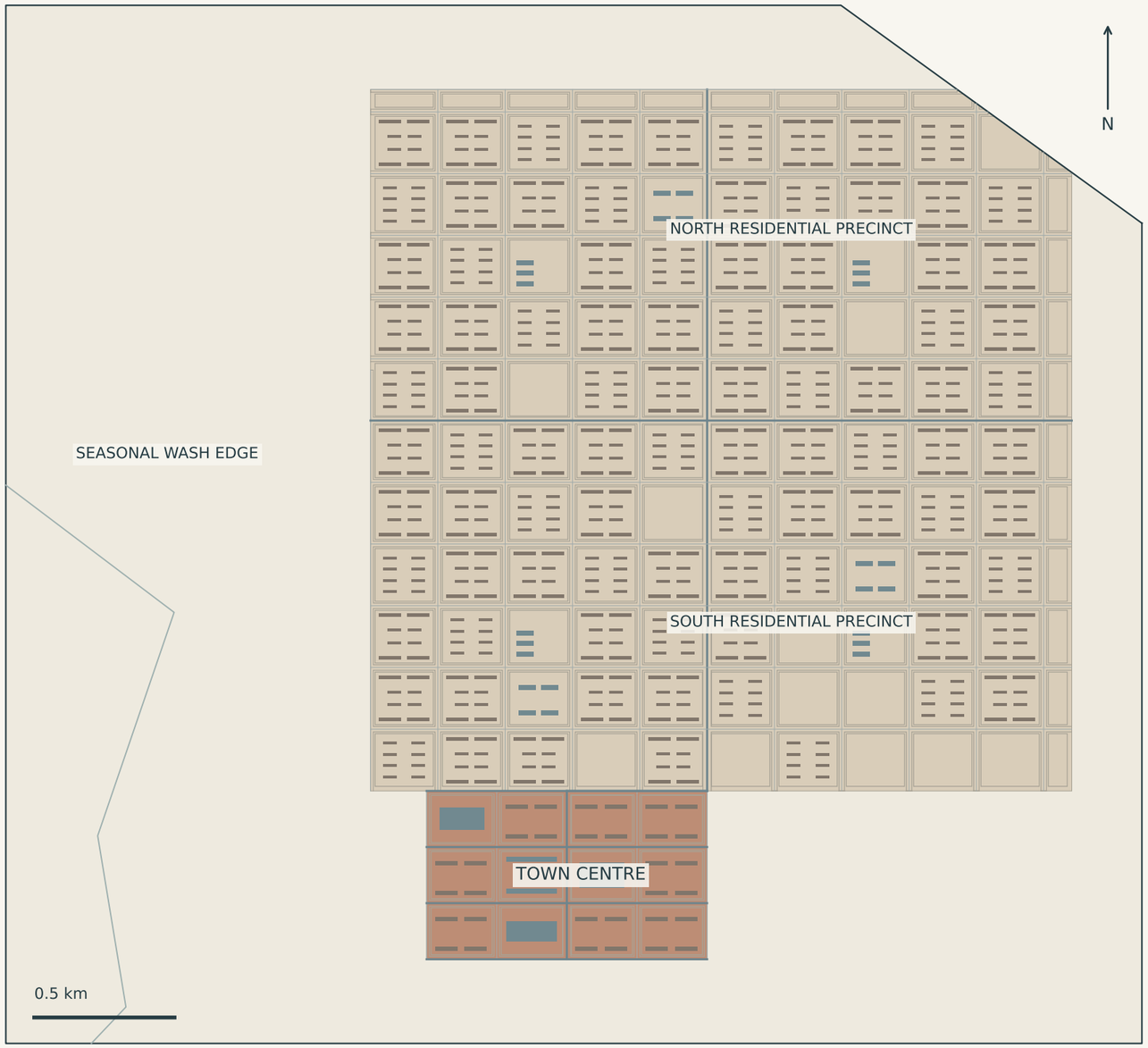
P + S / separate productive system



Housing
Mixed use
School
Daily services
Dry park
Compute
Workshops
Solar
Terrain reserve

01 / Planning structure

One residential district with connected release precincts, a town-level centre and separate production; no repeated micro-city packages.



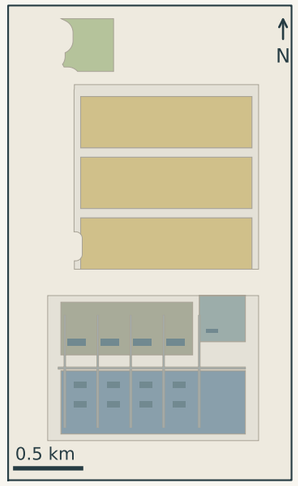
Decision / keep housing continuous

Secondary streets connect the south and north precincts. Services and open-space blocks reinforce the residential fabric. There is no compute campus, logistics yard or institutional barrier inserted between them.

Decision / scale centres by role

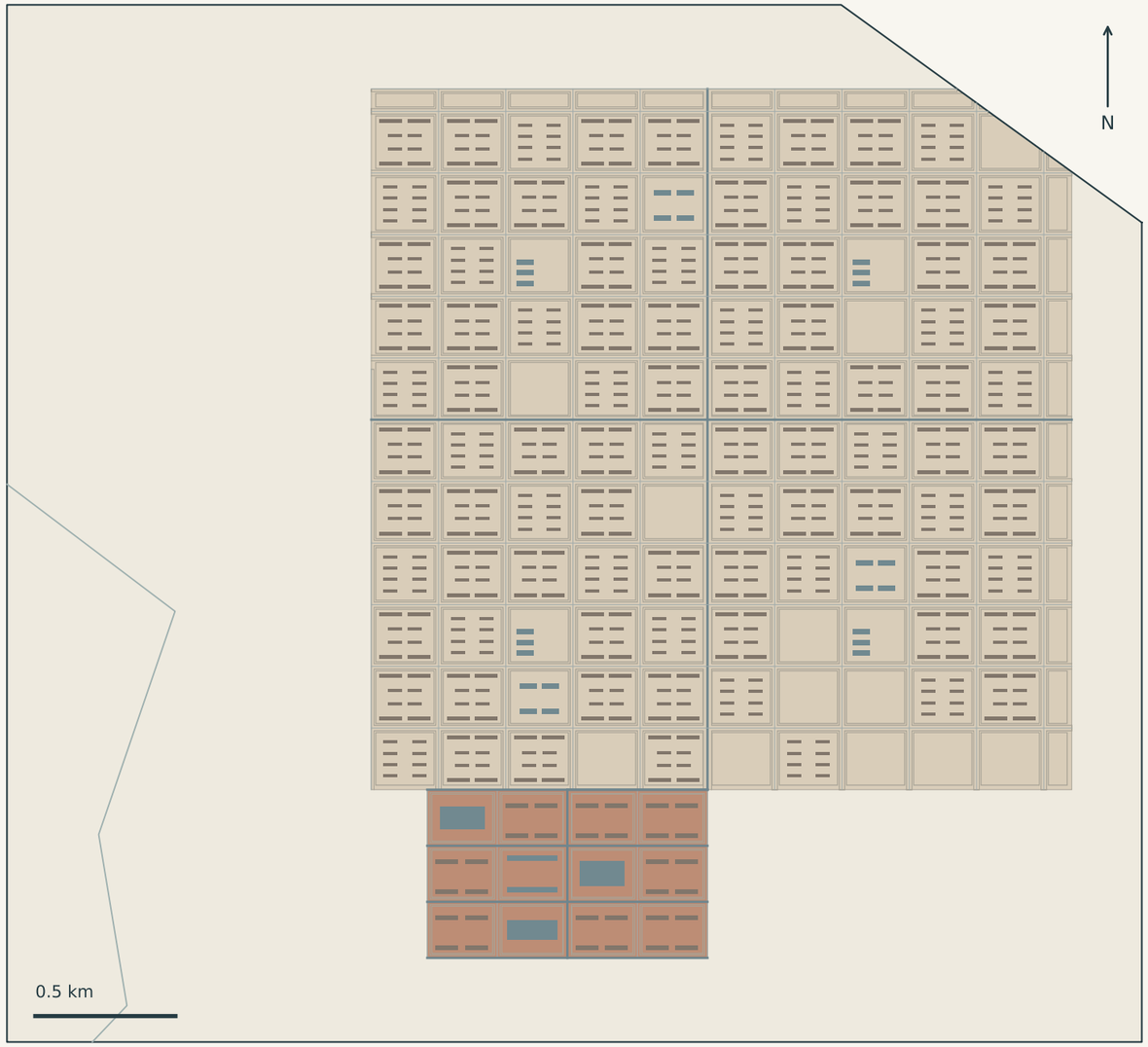
Daily centres support their surrounding homes. C serves the wider town. P/N/E serve productive systems with their own access and operating arrangements. More distance by itself earns no design credit.

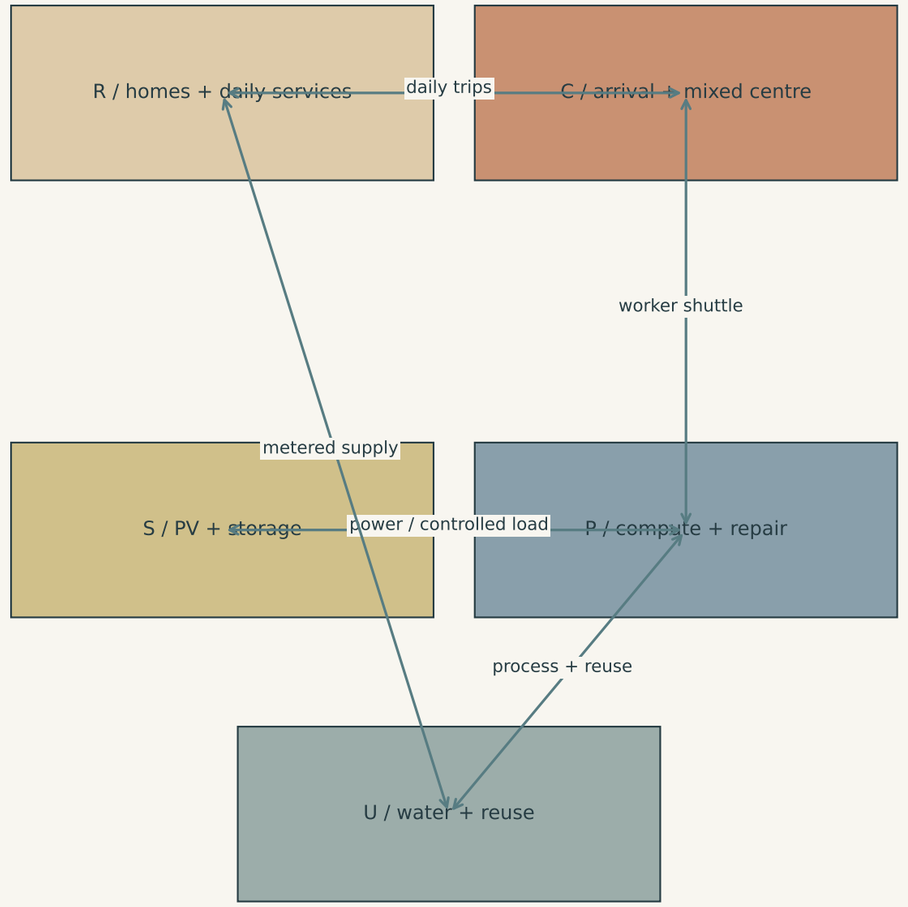
P + S / separate productive system



02 / Functional programme and operating links

Functions are relationships between activities; the separate land-use sheet accounts for spatial units and areas.

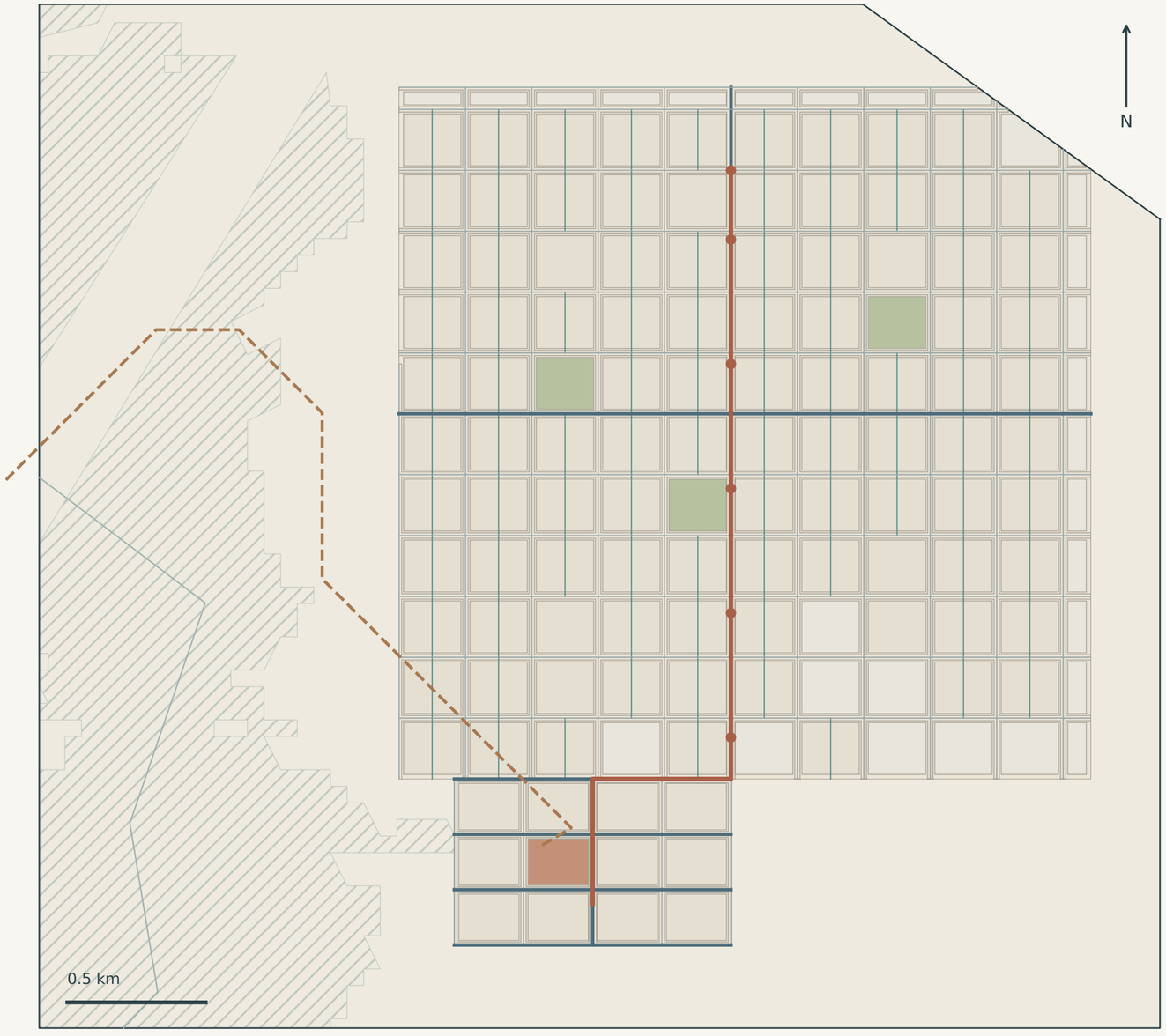




No residential heat pipe is credited. Production heat reuse requires a real nearby receiver and tested ten

03 / Streets, mobility and arrival

Connected primary/secondary/local streets, public transport and court access; regional and production freight are separate corridor studies.



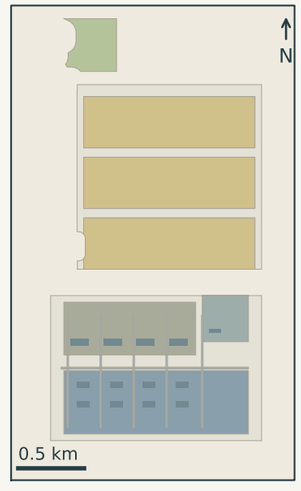
Town movement hierarchy

Primary arrival meets C. Secondary distributors and a bus spine connect residential precincts. Local streets and public court paths reach individual blocks; school entrances are served by local routes, not production freight.

Associated corridors are measured studies

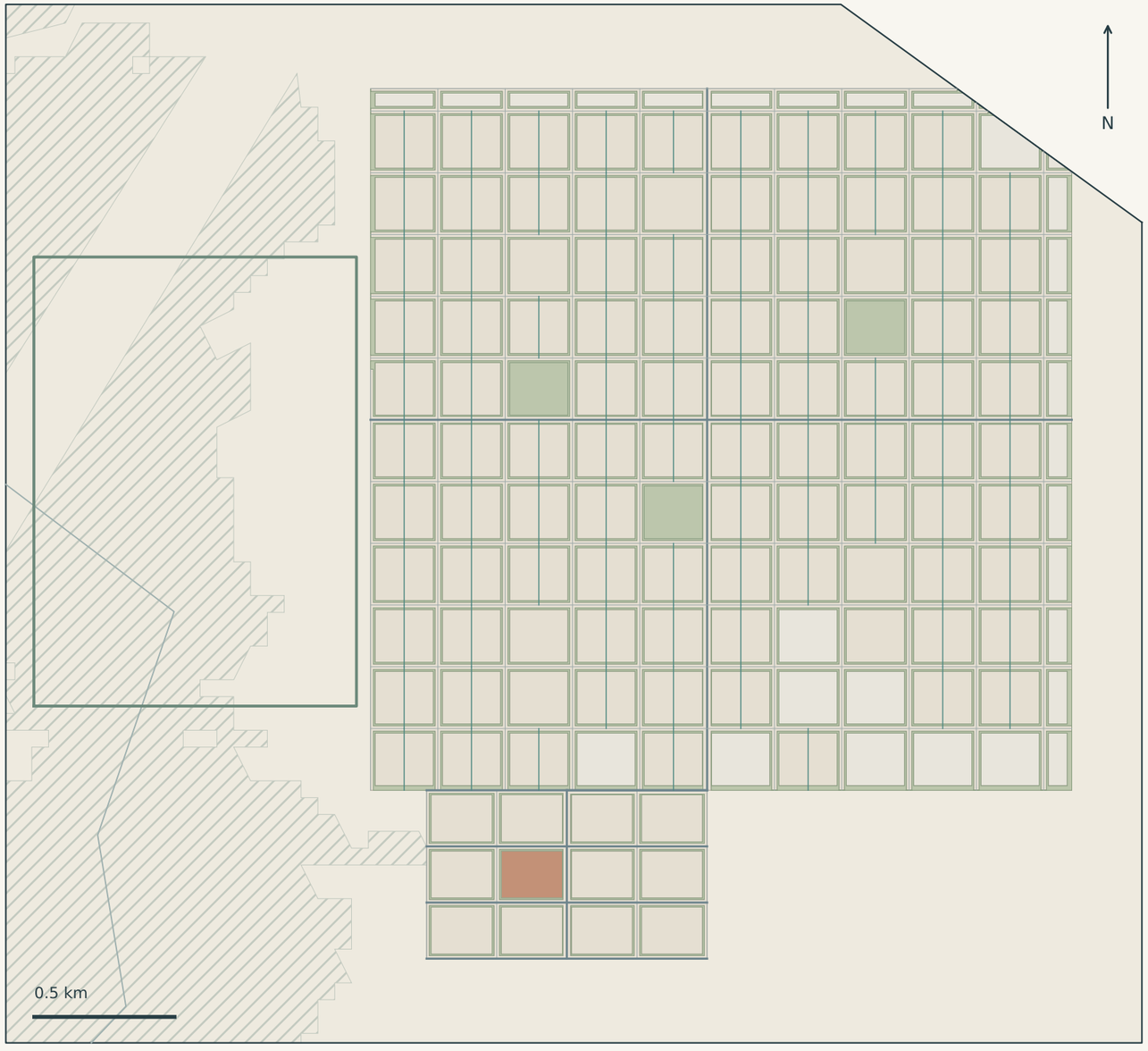
F bypasses R and C; the worker shuttle connects C and P. RN40 access and bridge/culvert designs remain unresolved enabling works. A drawn line does not establish an acquired right or an operating road.

P + S / separate productive system



04 / Ecology, landscape and water-sensitive public space

Landscape connects existing terrain/channel structure to dry public space and engineered shade; flood screening is not a lake.



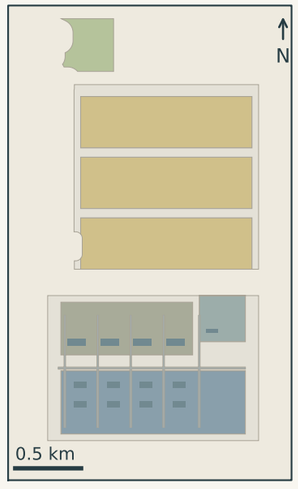
Source landscape and designed ground

Mapped washes retain conveyance; C03 visitor paths are removable and close on warning. The eastern edge has terrain-review land. Neither area is treated as a verified hydraulic boundary or as ornamental water.

Shade does not mean irrigated forest

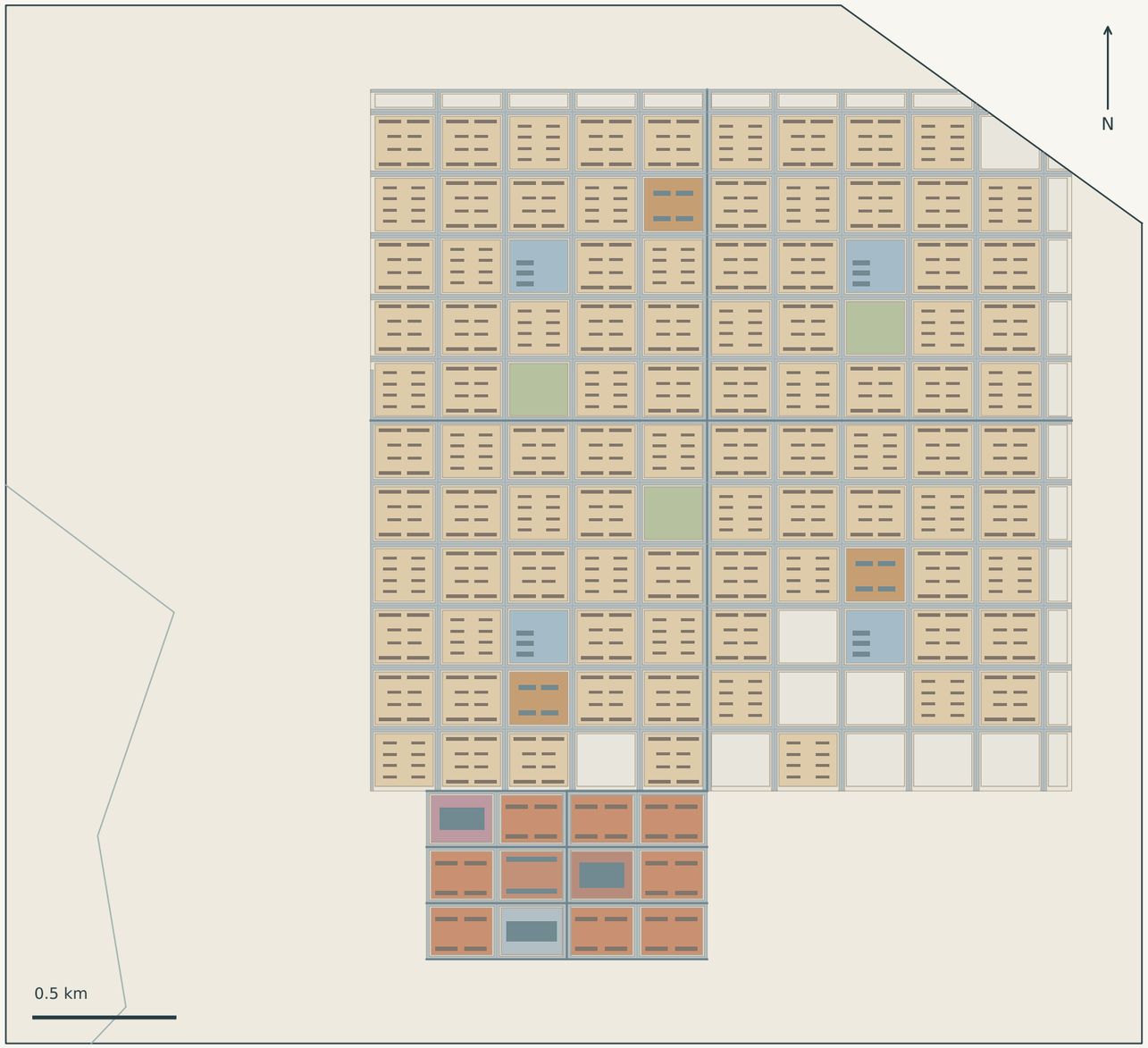
Public court routes, arcades and roofed stops provide explicit shade opportunities. Dry parks use mineral ground and limited planting subject to reuse supply. No procedural trees, canals or lawns are inferred.

P + S / separate productive system



05 / Land-use units and reconciled accounts

Districts, functional activities, net land-use units and construction phases are different accounts, linked by stable IDs.



NET LAND / hectares

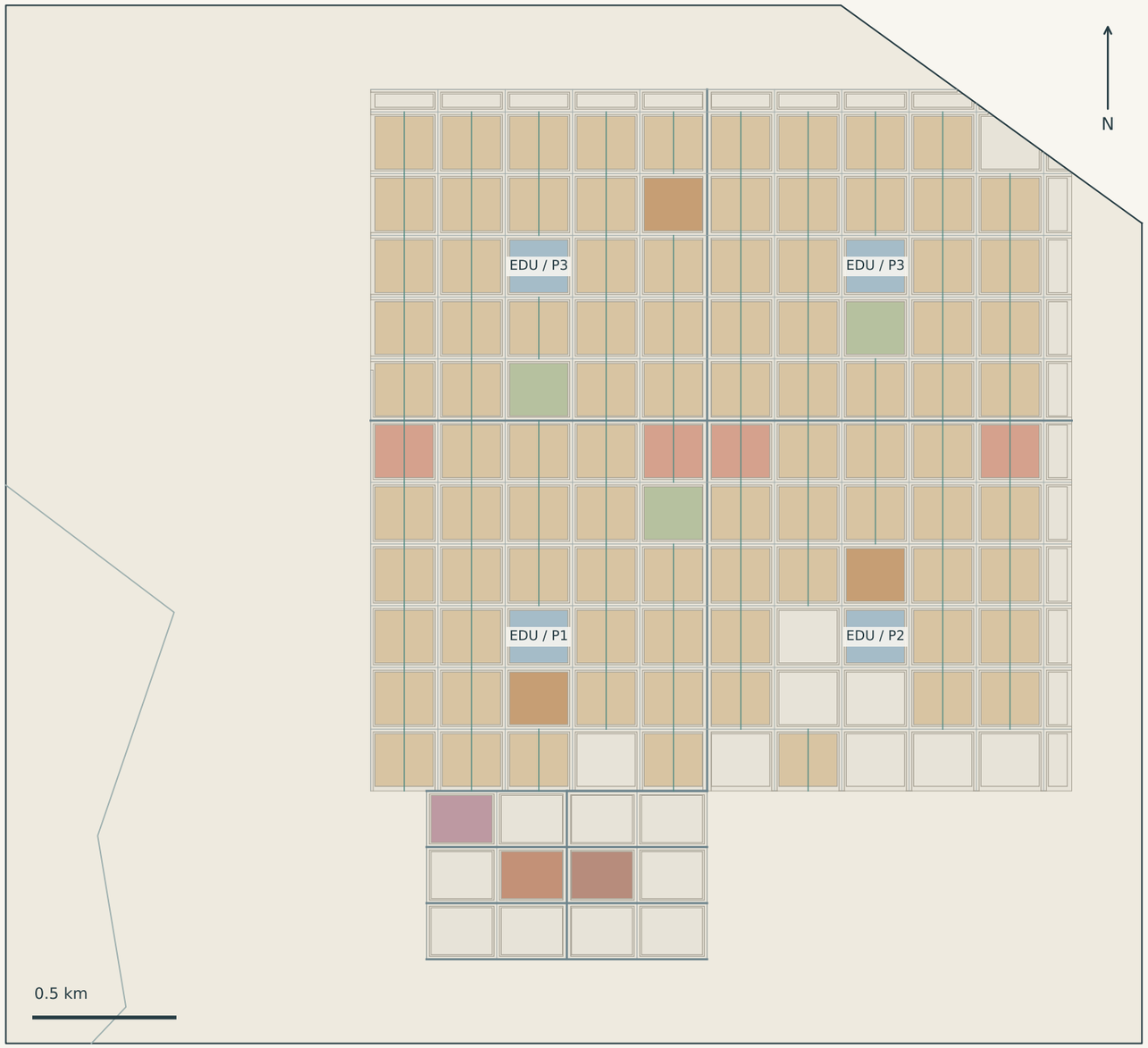
residential	355.8
operational reserve	346.8
energy	151.9
streets	130.7
dry open space	66.5
compute	64.6
terrain review	56.2
industry	40.6
mixed use	29.3
school	15.6
agriculture	13.4
infrastructure	12.1
civic	12.1
daily centre	11.7

All seven district accounts close without overlap or omission. Open and terrain-review ground has no residential capacity. Floor area is computed from actual submitted footprints x floors.

Housing form is a design scenario: 120 m2 gross residential floor area per dwelling. GFA, dwellings, school provision and phase quantities share the same building IDs.

06 / Public space, education and daily services

Capacity and access are checked together: this is a block-gate network test, not an unverified door-to-door walkability claim.



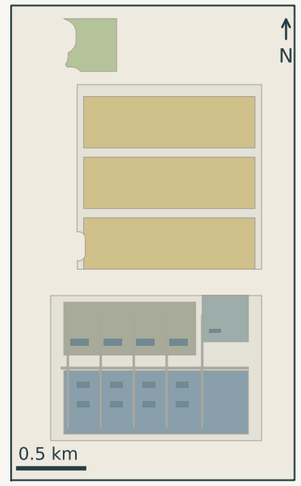
Education grows with housing

Full-city resident scenario 25,600;
education demand at assumed 15% is 3,840.
Designed school capacity is 4,320 places.
The age model and local operating
standards require verification.

Measured street-network coverage

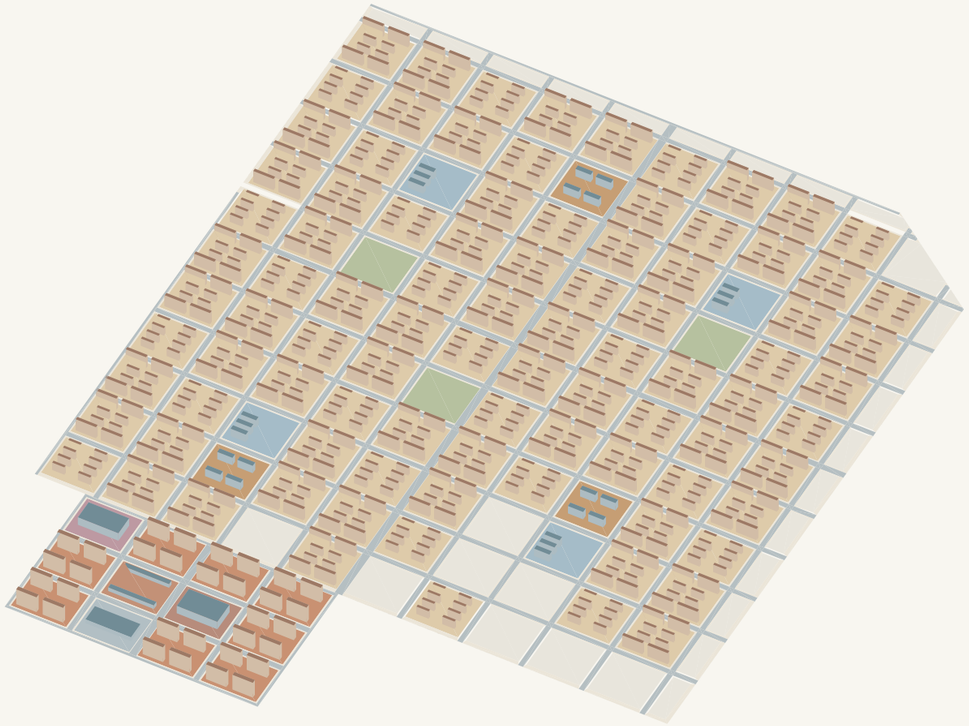
Full-layout median block-gate distance 680 m; maximum 1140 m. 4 blocks exceed the 1,000 m review marker. Those routes need safe crossings and school transport; the marker is not a legal standard.

P + S / separate productive system



Overall spatial view from the same complete layout

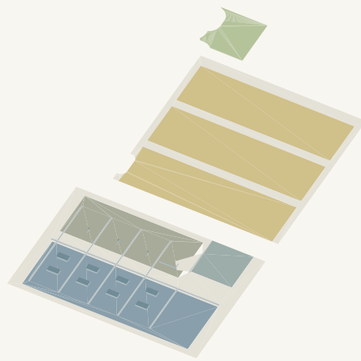
Diagrammatic orthographic axonometric / full build-out / flat analytical datum; building heights x3 for legibility, not terrain or a physical camera.



The urban whole is visible

Every drawn housing block, town-centre building and school is taken from the same geometry as the plans and accounts. The main frame covers R and C, not the full competition site.

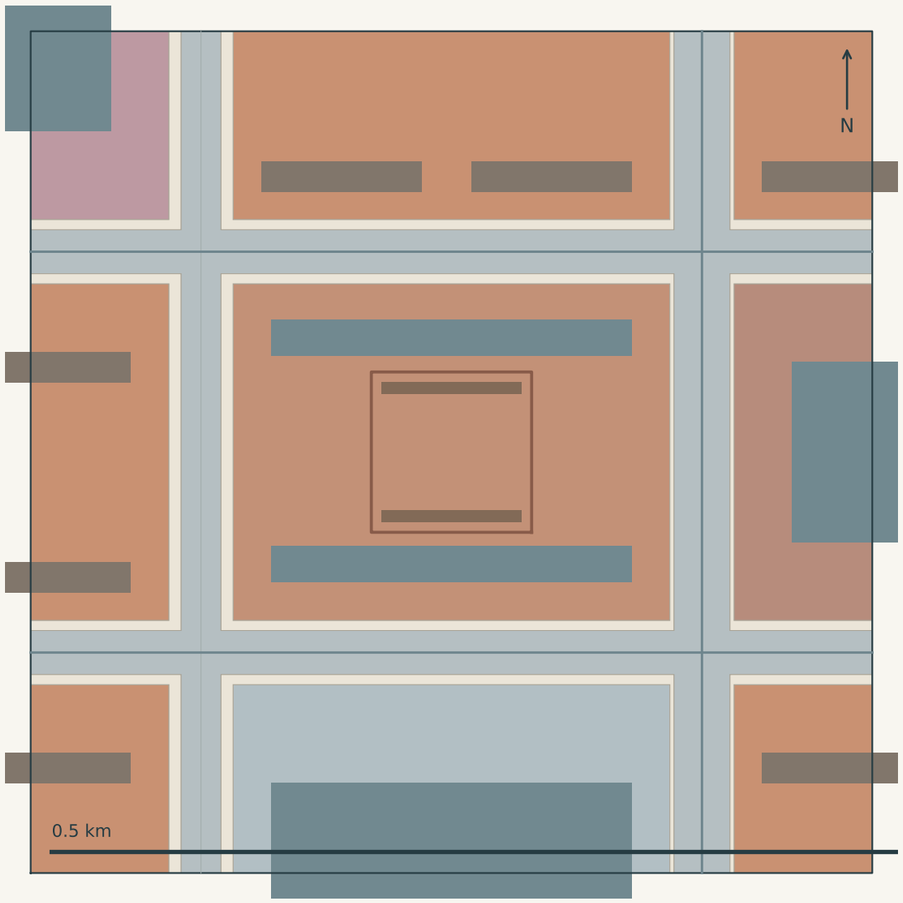
Separate P + S inset / same orientation

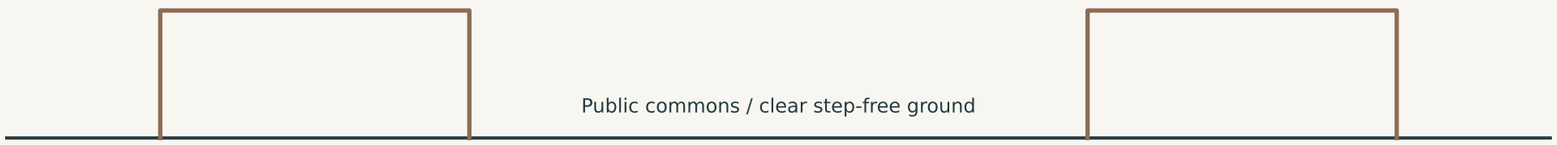


P is not moved beside R to fit this sheet. The full-site locator shows N and E; their workplace plans are accounted separately. No local aerial is presented as the whole territory.

A03 / Market commons and climate refuge

Node extent and phase are independent of the urban-design enlargement. Plans and sections refer to the current geometry.





City promise tested

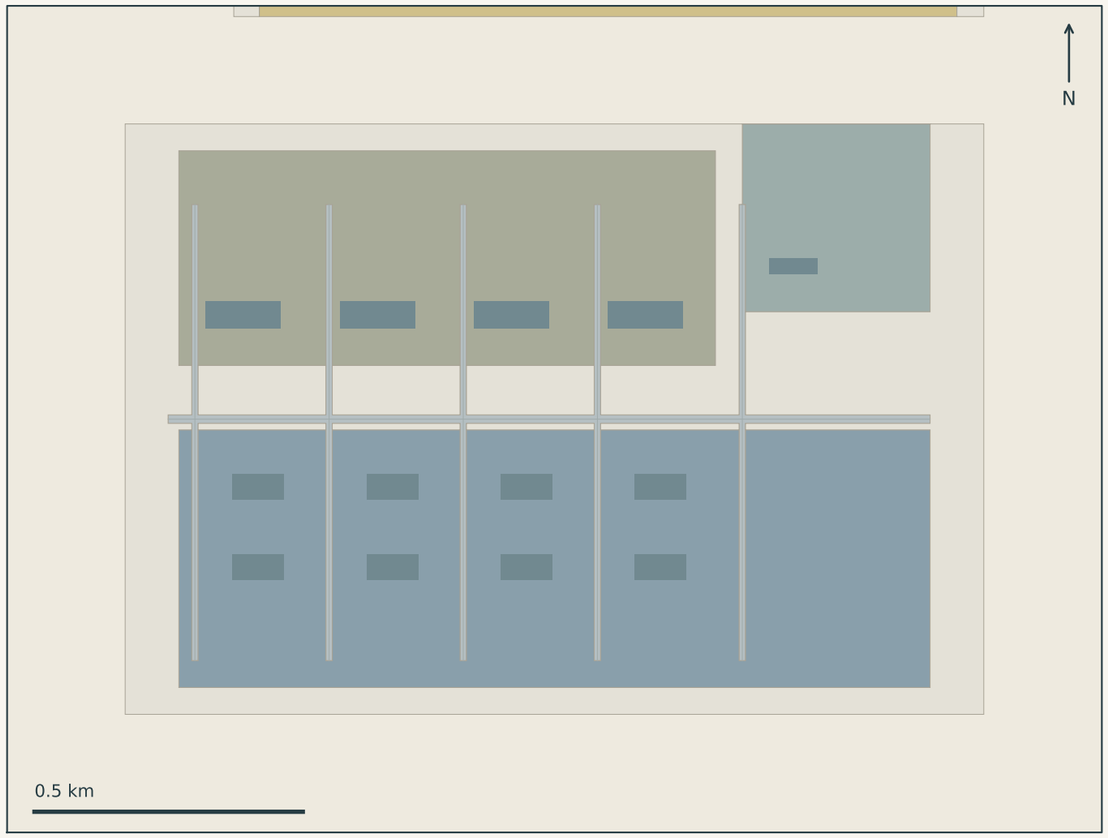
A 0.64 ha commons belongs to the town-level market/refuge, not to every housing block. Two canopy bands, shaded edges and nearby indoor refuge support public use. Seasonal shade, structure and operational staffing remain to verify.

Current view status

These are formal source plans and explanatory sections. New finished aerial and eye-level perspectives must later use matching controls and registered camera footprints. Older node images are not used for this redesign.

B03 / Compute and repair precinct

Node extent and phase are independent of the urban-design enlargement. Plans and sections refer to the current geometry.



City promise tested

Compute and repair share a controlled productive precinct with water metering and dry cooling. Housing remains in R. No long residential heat network is credited. Phase 1 is 2 MW; expansion reaches 8 MW only after resource and demand gates.

Current view status

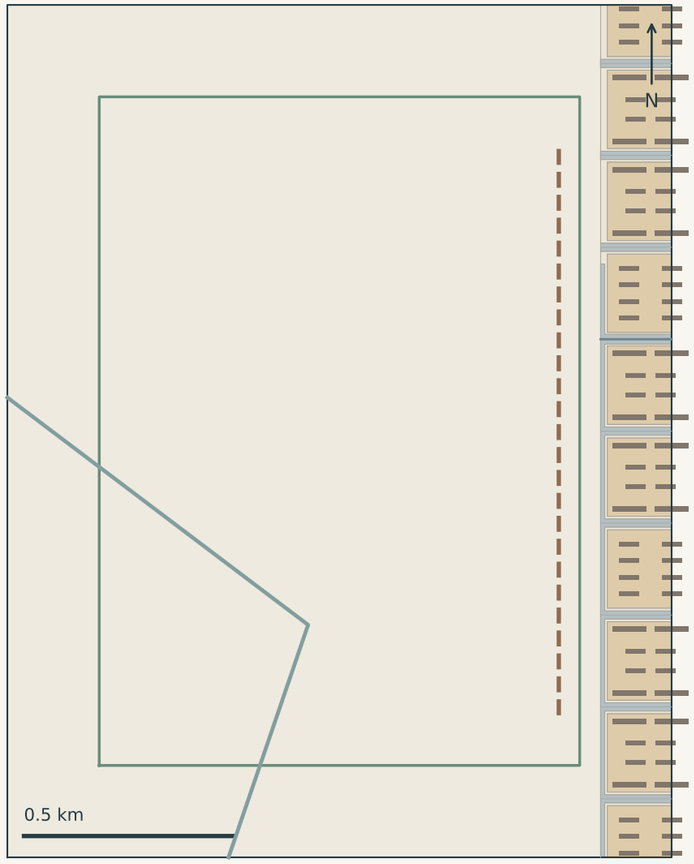
These are formal source plans and explanatory sections. New finished aerial and eye-level perspectives must later use matching controls and registered camera footprints. Older node images are not used for this redesign.

Concept hall section / 8 MW in eight staged 1 MW halls



C03 / Seasonal wash park

Node extent and phase are independent of the urban-design enlargement. Plans and sections refer to the current geometry.



Diagrammatic wash section / not surveyed flood levels



City promise tested

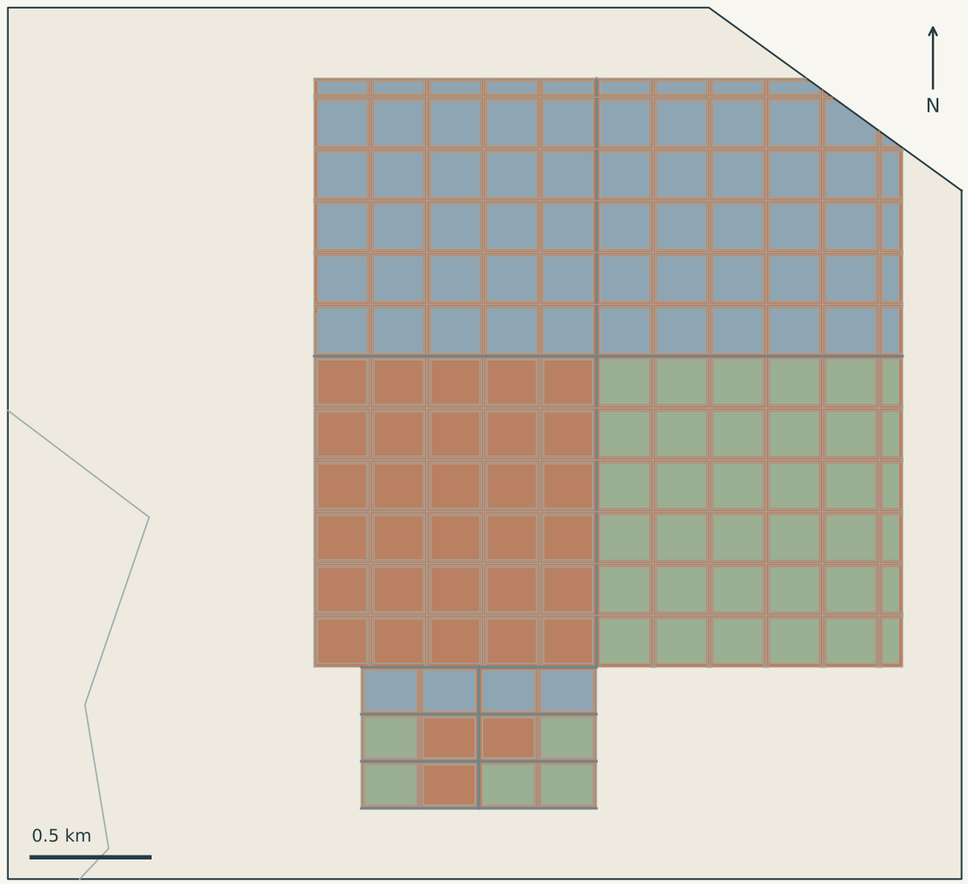
The source channel is a seasonal conveyance route, not a lake. The visitor path stays on the edge, is removable and closes on warning. No occupied pavilion or dam is placed on the channel. Actual safe edges depend on survey and hydraulics.

Current view status

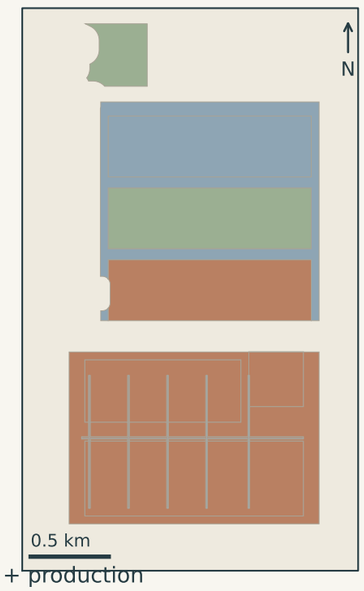
These are formal source plans and explanatory sections. New finished aerial and eye-level perspectives must later use matching controls and registered camera footprints. Older node images are not used for this redesign.

Three construction phases with matching capacity accounts

Every proposed development zone appears in the same geometry-linked programme; timing is conditional, not an approved calendar.



Separate productive phases



P1: core life + production
P2: connected district expansion
P3: northern fabric + gateway

N and E have their own service, transport and utility conditions.

Phase	New people	Total people	New homes	Total homes	New jobs	Total jobs
1	7,040	7,040	2,816	2,816	783	783
2	6,440	13,480	2,576	5,392	1,866	2,649
3	12,120	25,600	4,848	10,240	1,812	4,461

Resources, livelihoods and independent first operation

Demand is computed before claiming supply. Missing rights and fixed-pilot demand are not treated as zero.

Phase 1 / a measured demand scenario

7,040 residents; 783 design job capacity. Fresh source requirement for the NEW programme: 409,590 m3/year. Existing fixed pilots/quarter remain additional and unverified.

Synthetic hourly dispatch

16 MWp PV / 40 MWh battery / 2 MW peak IT. Fully matched hours: 82.52%; achieved IT utilisation: 70.4%. Unmatched grid imports are not credited carbon-free. No power export is assumed.

First works and independent operation

Regional access and corridor rights; verified source/yield/quality; covered storage and reuse; power/fibre and protected essential loads; school/health/market operators; wastewater/waste and emergency response. Housing occupancy waits for these, even if later stages stop.

A 30% water shortage changes use

Protect drinking water, care, wastewater and essential safety. Curtail new irrigation trials and discretionary cleaning, then interruptible production/IT. Existing pilot obligations need operator agreement; their water is not silently confiscated.

Livelihood and demand are real gates

Employment totals are design capacity, not signed jobs. The workforce/income mix and the demand for occupied homes require an evidenced case. Release housing in funded tranches rather than constructing the entire capacity scenario at once.

Checks and remaining verification

815 buildings; 91 housing blocks. Land accounts close and encoded building/road overlaps are checked. These checks do not establish water rights, school operation, highway access, seismic design, noise performance or image approval.